

DSA0502 – Query Processing for Data Science

PRACTICE PROGRAMS

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Date: 30/07/2026

Program 1: Password Validation using Regular Expressions

Question

Write a Python program to validate a password based on the following criteria:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one digit

Aim

To validate a password using Python Regular Expressions.

Code

```
import re

password = input("Enter Password: ")

pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d).{8,}$'

if re.match(pattern, password):

    print("Valid Password")

else:

    print("Invalid Password")
```

Sample Output

Output 1

Enter Password: Hello123

Valid Password

Output 2

Enter Password: hello12

Invalid Password

```
import re

password = input("Enter Password: ")

pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d){8,}$'

if re.match(pattern, password):
    print("Valid Password")
else:
    print("Invalid Password")
|

... Enter Password: SWEETY
Invalid Password
```

Result

Thus, the Python program to validate a password using Regular Expressions was executed successfully.

Program 2: Extract URLs using Regular Expressions

Question

Write a Python program to extract all URLs from a given paragraph using Regular Expressions.

Aim

To extract all URLs from a paragraph using Python Regular Expressions.

Code

```
import re

text = """

Visit https://www.google.com for searching.

Our website is https://www.python.org.

Also visit http://example.com.

"""

pattern = r'https?://[^\s]+'
urls = re.findall(pattern, text)

print("URLs Found:")

for url in urls:
    print(url)
```

Sample Output

```
import re
text = """
Visit https://www.google.com for searching.
Our website is https://www.python.org.
Also visit http://example.com.
"""

pattern = r'https?:\/\/[^\s]+'
urls = re.findall(pattern, text)
print("URLs Found:")
for url in urls:
    print(url)
```

... URLs Found:
<https://www.google.com>
<https://www.python.org>
<http://example.com>

Result

Thus, the Python program to extract URLs using Regular Expressions was executed successfully.

Program 3: Validate Indian PIN Code

Question

Write a Python program to validate a six-digit Indian PIN code using Regular Expressions.

Aim

To validate an Indian PIN code using Python Regular Expressions.

Code

```
import re

pin = input("Enter PIN Code: ")

pattern = r'^[1-9][0-9]{5}$'

if re.match(pattern, pin):
    print("Valid PIN Code")
else:
    print("Invalid PIN Code")
```

Sample Output

```
import re
pin = input("Enter PIN Code: ")
pattern = r'^[1-9][0-9]{5}$'
if re.match(pattern, pin):
    print("Valid PIN Code")
else:
    print("Invalid PIN Code")

... Enter PIN Code: 600102
Valid PIN Code
```

Result

Thus, the Python program to validate an Indian PIN code using Regular Expressions was executed successfully.

Program 4: Extract Currency Values

Question

Write a Python program to extract all currency values from a given text using Regular Expressions.

Aim

To extract currency values from text using Python Regular Expressions.

Code

```
import re

text = """

The laptop costs ₹55000.

The mobile costs $899.

The watch costs ₹3500.

The bag costs $120.

"""

pattern = r'[₹$]\d+(?:\.\d+)?'

currency = re.findall(pattern, text)

print("Currency Values:")

for value in currency:
    print(value)
```

Sample Output

```

import re
text = """
The laptop costs ₹55000.
The mobile costs $899.
The watch costs ₹3500.
The bag costs $120.
"""

pattern = r'[₹$]\d+(?:\.\d+)?'
currency = re.findall(pattern, text)
print("Currency Values:")
for value in currency:
    print(value)

```

```

... Currency Values:
₹55000
$899
₹3500
$120

```

Result

Thus, the Python program to extract currency values using Regular Expressions was executed successfully.

Program 5: Extract Dates (DD/MM/YYYY)

Question

Write a Python program to extract all dates in the format DD/MM/YYYY from a given text using Regular Expressions.

Aim

To extract dates in DD/MM/YYYY format using Python Regular Expressions.

Code

```

import re

text = """
Exam Date: 15/08/2026
Submission Date: 20/09/2026
Meeting Date: 01/01/2027
"""

pattern = r'\b\d{2}/\d{2}/\d{4}\b'
dates = re.findall(pattern, text)

```

```
print("Dates Found:")
```

```
for date in dates:
```

```
    print(date)
```

Sample Output

```
import re
text = """
Exam Date: 15/08/2026
Submission Date: 20/09/2026
Meeting Date: 01/01/2027
"""

pattern = r'\b\d{2}/\d{2}/\d{4}\b'
dates = re.findall(pattern, text)

print("Dates Found:")
for date in dates:
    print(date)
```

```
... Dates Found:
15/08/2026
20/09/2026
01/01/2027
```

Result

Thus, the Python program to extract dates in DD/MM/YYYY format using Regular Expressions was executed successfully.